

Sales Data Analysis

Name : Tanya Adhikari

Intern ID : ITID4469

Project Title : Sales Data Analysis

Tool Used : Power BI

Dataset : Walmart Sales Dataset

Introduction

Sales data analysis plays a crucial role in understanding business performance and customer demand patterns. With the growing availability of large datasets, organizations rely on data analytics tools to gain meaningful insights and support decision-making.

This project focuses on analyzing Walmart Sales Data using Power BI to identify sales trends, seasonal patterns, store-wise performance, and the impact of holidays and external factors. The insights obtained can help management improve sales forecasting, inventory management, and promotional strategies.

Objective of the Project

The main objectives of this project are:

- To analyze sales data and identify trends and seasonality
- To evaluate store-wise sales performance
- To study the impact of holidays on sales
- To understand the relationship between external factors (temperature, CPI, unemployment) and sales
- To provide data-driven recommendations for improving sales and inventory planning

Exploratory Data Analysis

Exploratory Data Analysis was conducted using Power BI to gain deeper insights into sales patterns and influencing factors. Store-wise analysis revealed significant variation in total sales, indicating differences in store performance and customer demand. Monthly sales trends highlighted clear seasonality, with sales peaking during specific months each year. Year-wise comparison showed consistent demand patterns, making sales behavior predictable over time.

The analysis of average weekly sales based on the holiday flag confirmed that sales increase during holiday periods, reflecting higher consumer spending. Additionally, the relationship between temperature and total sales showed a moderate correlation, suggesting that weather conditions influence purchasing behavior to some extent. External economic indicators such as CPI and unemployment were also reviewed to understand their impact on sales. Overall, EDA helped identify key trends, seasonal patterns, and factors driving sales performance.

Dashboard Visuals and Explanation

- **VISUAL 1 : Total Sales by Store**

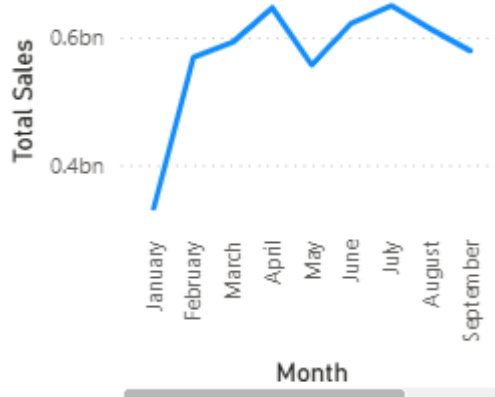
Total Sales by Store



This bar chart shows total sales for each store. It helps identify high-performing and low-performing stores. Stores with higher bars generate more revenue, while stores with lower bars indicate lower sales performance.

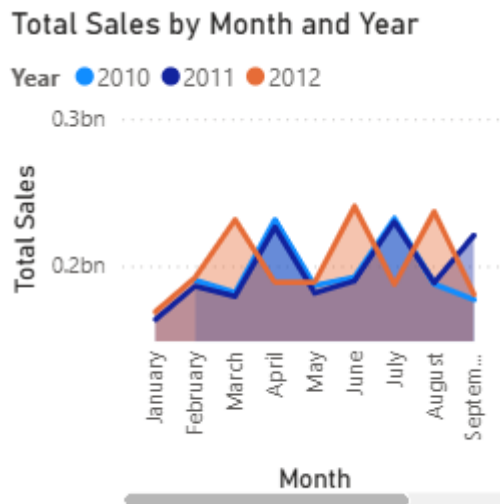
- **VISUAL 2 : Total Sales by Month**

Total Sales by Month



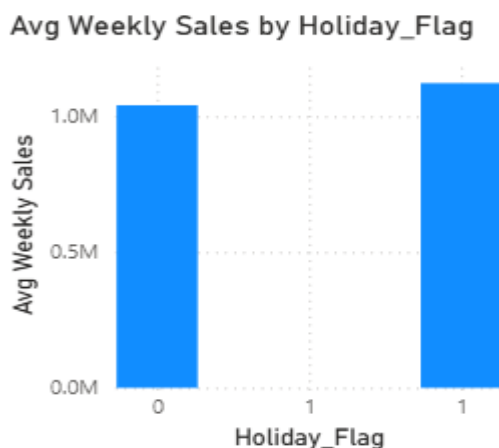
This line chart represents total sales across different months. It highlights seasonal trends in sales. Sales increase during certain months, indicating higher customer demand during those periods.

- **VISUAL 3 : Total Sales by Month and Year**



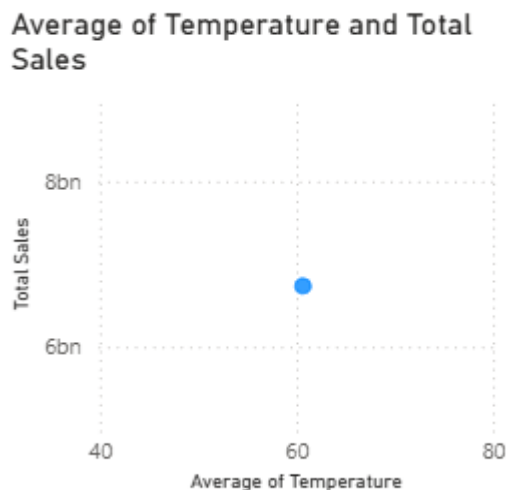
This visualization compares monthly sales across different years. It helps identify year-over-year trends and sales growth patterns. Variations in sales indicate seasonal and yearly performance changes.

- **VISUAL 4 : Average Weekly Sales by Holiday Flag**



This chart compares average weekly sales during holiday and non-holiday periods. It shows that sales are generally higher during holidays, indicating the positive impact of holidays on customer purchasing behavior.

- **VISUAL 5 : Average Temperature vs Total Sales**



This scatter plot shows the relationship between average temperature and total sales. It helps analyze whether temperature has an impact on sales. The visualization indicates that sales vary with temperature, but other factors also influence sales performance.

High and Low Performing Stores

High-performing stores are those that generate the highest total sales. These stores consistently show strong sales performance and contribute significantly to overall revenue.

Low-performing stores generate comparatively lower sales. These stores may be affected by factors such as location, lower customer demand, or operational challenges. Identifying these stores helps management take corrective actions to improve performance.

Feature

Key features affecting sales performance include store number, date, holiday flag, temperature, and economic factors such as fuel price and unemployment rate. Among these, holidays and seasonal trends have a major impact on sales.

Findings

1. Sales Performance Across Stores

- Sales are **unevenly distributed across stores**.
- A few stores contribute a **significantly higher share of total sales**, while some stores consistently underperform.
- This indicates differences in **store location, customer demand, and operational efficiency**.

2. Monthly and Seasonal Trends

- Sales show a **clear seasonal pattern**, with noticeable peaks during certain months.
- Higher sales are observed in **mid-year and festive months**, indicating strong seasonality.
- Monthly trends remain consistent across years, suggesting **predictable demand cycles**.

3. Impact of Holidays

- Average weekly sales are **higher during holiday periods** compared to non-holiday weeks.
- This confirms that **holidays positively influence customer spending behavior**.

4. Effect of Temperature and External Factors

- Temperature shows a **moderate relationship with sales**, implying weather conditions influence purchasing patterns.
- External economic indicators (like CPI and unemployment) may also affect sales but are **secondary compared to seasonality and holidays**.

Recommended Strategies

1. Improve Inventory Planning

- Increase inventory levels **before peak and holiday seasons** to avoid stockouts.
- Reduce excess inventory during low-demand months to **minimize holding costs**.

2. Store-Specific Strategies

- Analyze high-performing stores to identify **best practices**.
- Apply similar pricing, product mix, and promotional strategies to **low-performing stores**.

3. Holiday-Focused Promotions

- Launch **targeted marketing campaigns and discounts** during holidays.
- Ensure sufficient staffing and product availability during these periods to maximize sales.

4. Seasonal Demand Forecasting

- Use historical sales patterns to build **accurate sales forecasts**.
- Align procurement and logistics planning with predictable seasonal demand.

5. Data-Driven Decision Making

- Regularly monitor KPIs such as **weekly sales, store performance, and holiday impact** using Power BI dashboards.
- This enables management to make **quick and informed business decisions**.

Conclusion

The Sales Data Analysis project using Walmart sales data successfully demonstrated how business intelligence tools like Power BI can be used to transform raw data into meaningful insights. The analysis revealed clear patterns in sales behavior across different stores, months, and years. Store-wise analysis showed that sales performance varies significantly, highlighting the need for customized strategies for different locations. Seasonal trends were clearly visible, with sales increasing during specific months and festive periods, making demand patterns predictable.

The impact of holidays was found to be a major driver of sales, as average weekly sales were consistently higher during holiday periods. External factors such as temperature and economic indicators showed a moderate influence on sales but were less impactful compared to seasonality and holidays. Overall, the project emphasizes the importance of data-driven decision-making in improving sales forecasting, inventory management, and strategic planning, ultimately helping organizations enhance operational efficiency and profitability.